

Abubakar Diallo

New York City, NY | dialoabo@gmail.com | 347-824-0903 | [LinkedIn](#) | [GitHub](#)

EDUCATION

New York University

B.S. Computer Science — New York City, NY

Expected May 2027

Borough of Manhattan Community College

A.S. Computer Science — New York City, NY

Aug 2023 – Jun 2025

RESEARCH EXPERIENCE

Honors Research Projects — Borough of Manhattan Community College

Spring 2025

Undergraduate Researcher | CS Honors Program

- Conducted empirical research on the computational significance of zero in binary systems and data structures (CSC 331H); produced a formal [research paper](#) and delivered a [presentation](#) — demonstrating technical writing and research methodology skills aligned with AI safety documentation workflows.
- Built a Qt/C++ desktop application for CS education with user authentication, course catalog, and integrated learning resources (CSC 211H), applying end-to-end software engineering and system design practices. [\[GitHub\]](#) [\[Video\]](#)
- Developed skills in research methodology, academic writing, and communicating complex technical findings to non-specialist audiences — directly applicable to producing public research outputs (papers, reports).

EXPERIENCE

Break Through Tech Fellow — Cornell Tech @ NYC

May 2025 – Present

AI/ML Fellowship · Selected from 1,500+ applicants

- Participating in a competitive 12-month AI/ML program combining ML coursework, industry mentorship, and applied AI research projects — directly mirroring the Anthropic Fellows learning structure.
- Developing hands-on experience with large language models, ML pipelines, and empirical AI research methods across industry-partnered projects.

AI Studio Apprentice — Estée Lauder (via Break Through Tech) [\[GitHub\]](#)

Sep – Dec 2025

Industry ML Research Project

- Led EDA, feature engineering, and model evaluation within a 7-member cross-functional team to deliver a client-ready AI prototype using Python and scikit-learn.
- Applied MLRATE estimator to quantify randomized intervention impact; documented technical approach, use cases, and model outcomes for stakeholder review — mirroring agentic AI documentation and evaluation workflows.
- Communicated research findings clearly to non-technical stakeholders, translating ML outputs into actionable insights. [\[Presentation\]](#)

Web Developer — STEMKasa

Jul – Aug 2024

New York City, NY

- Built a responsive React/HTML/CSS frontend for an AI-powered student chatbot integrated with the Mistral AI API.
- Engineered prompt flows to enable real-time, contextually accurate student inquiry responses via LLM API calls.
- Collaborated with a 3-person team to ensure seamless UX across interactive AI-driven response cycles.

PROJECTS

Gando AI — African-Language-First Agentic App Builder [\[GitHub\]](#)

Apr 2026

React, TypeScript, Node.js/Express, Gemini 2.5 API, Firebase

- Built a Replit/Lovable-style agentic AI app builder where users describe web apps in African languages (Fulani ADLaM, Swahili, Yoruba, Hausa) and receive working HTML/CSS/JS output with explanations in their native language.
- Designed a single-call Gemini 2.5 pipeline that auto-detects input language, generates a full web app, and writes an explanation in one structured JSON response — eliminating costly translation round-trips.
- Implemented iterative chat-driven refinement with full version history and one-click revert, simulating agentic helpdesk-style workflows for non-English users.
- Integrated Firebase Auth and Firestore persistence; deployed with Vite + Express proxy to protect API keys.

Country Happiness Score Predictor — Break Through Tech [\[GitHub\]](#)

May – Jul 2025

Python, scikit-learn, pandas, matplotlib

- Built an ML pipeline to predict national happiness scores using the World Happiness Report dataset with EDA and feature engineering.
- Trained a Random Forest regression model achieving 86.17% accuracy, outperforming linear baselines by 12% via GridSearchCV hyperparameter tuning.
- Visualized GDP per capita and health metrics as primary happiness drivers for policy-level insights.

SkillSpire LMS — Full-Stack Learning Management System [\[GitHub\]](#)

Dec 2024 – Mar 2025

React, Node.js, Express.js, MongoDB, Bootstrap, JWT

- Developed a full-stack MERN application with JWT-based authentication, RESTful APIs, course enrollment, and tutor-student matching.
- Built responsive React frontend with Bootstrap 5, real-time EmailJS integration, and mobile-optimized UI components.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, C++, HTML/CSS

AI / LLM: OpenAI API, Claude (Anthropic) API, Gemini API, Mistral AI API, LangChain, Prompt Engineering, RAG, large language models, mechanistic interpretability (foundational)

ML / Data Science: scikit-learn, pandas, NumPy, matplotlib, GridSearchCV, feature engineering, model evaluation, EDA, Jupyter

Frameworks: React, Node.js, Express.js, Flask, Bootstrap, Docker, Qt

Databases: MongoDB, PostgreSQL

Tools: Git/GitHub, Postman, VS Code, PyCharm, CMake, Firebase